UNI GRAZ

Complex Social &
Computational Systems

ML & PSYCHOLOGY IN THE LITERATURE

A PSYCHOLOGIST'S GUIDE TO MACHINE LEARNING

ZURICH, MARCH 16-20, 2026

GET TOGETHER IN GROUPS OF (+/-) THREE TO REVIEW ONE OF SIX PAPERS COMBINING ML AND PSYCHOLOGY.

ORGANIZATIONAL PSYCHOLOGY

Wu, L., Wang, D., & Evans, J. (2019). **Large teams develop and small teams disrupt science and technology**. *Nature*, *566*, 378-382.

<u>Keywords</u>: Database of scientific papers, document embeddings, Doc2Vec, neural networks, topic modeling

8 pages

CLINICAL PSYCHOLOGY

Sharma, A., Miner, A. S., Atkins, D. C., & Althoff, T. (2020). **A computational approach to understanding empathy expressed in text-based mental health support**. *EMNLP*, 5263-5276.

<u>Keywords</u>: Data annotation, natural language processing, classification, transformers

9 pages

AFFECTIVE SCIENCE / LINGUISTICS

Di Natale, A., Pellert, M., & Garcia, D. (2021). **Colexification networks encode affective meaning**. *Affective Science*, *2*, 99-111.

Keywords: Networks, translation databases, dictionary approaches

12 pages

DIFFERENTIAL PSYCHOLOGY

Rasmussen, S. H. R., Bor, A., & Petersen, M. B. (2024). **The offline roots of online hostility: Adult and childhood administrative records correlate with individual-level hostility on Twitter**. *PNAS*, *121*(44), e2412277121.

Keywords: Combination of online & offline data, word embeddings, Twitter data, data annotation

5 pages

PSYCHOLOGICAL METHODS

Wulff, D., & Mata, R. (2023). **Semantic embeddings reveal and address taxonomic incommensurability in psychological measurement**. *Nature Human Behavior*, *9*, 944-954.

Keywords: Psychological assessment scales, jingle-jangle fallacies, embeddings, semantic similarity

10 pages

COGNITIVE SCIENCES

Binz, M., et al. (2025). **A foundation model to predict and capture human cognition**. *Nature*, *644*, 1002-1019.

Keywords: Decision making, experimental data, large language models, brain scans

7 pages

LEADING QUESTIONS

1. What were the research questions?
2. What datasets did the authors analyze?
3. Which ML techniques did the authors use?
4. Which variables did the authors collect and how were they operationalized?
5. What were the main results of the study?
6. What did you (not) like about the paper?
7. What questions does the paper leave open?

A FEW FINAL NOTES

- You can find all papers and a template with the leading questions online. The documents are linked in the GitHub repository of the course.
- You will most likely not have time to study the paper in depth and there is a high chance that you will not understand everything. Rather scan the paper for the most relevant information that you can grasp along the leading questions.
- If you have questions about certain expressions or machine learning vocabulary that you think are crucial to understand, feel free to ask me anytime!
- You can study the papers individually or discuss them together in your groups.

LITERATURE REVIEW UNTIL 12:30.
LUNCH BREAK AFTERWARDS.